

Impact of Input and Output Devices on a Virtual Ultrasound Training

Mareen Allgaier*
Otto-von-Guericke
University Magdeburg,
Germany
Research Campus
STIMULATE, Germany

Eric-Morten Dangszat
Otto-von-Guericke
University Magdeburg,
Germany

Florentine Huettl
University Medical Center
Mainz, Germany

Laura Isabel Hanke
University Medical Center
Mainz, Germany

Tobias Huber
University Medical Center
Mainz, Germany

Bernhard Preim
Otto-von-Guericke
University Magdeburg,
Germany

Christian Hansen†
Otto-von-Guericke
University Magdeburg,
Germany
Research Campus
STIMULATE, Germany



Figure 1: Three versions of an ultrasound training application with varying technical setup: a) haptic device with VR headset, b) haptic device with normal desktop, and c) mouse and desktop.

ABSTRACT

Performing ultrasound requires mental skills that have to be trained hands on. Virtual simulations can be employed to provide novice surgeons with a safe training environment prior to performing ultrasound on the patient. For both input and output there is a wide range of devices that are used in existing ultrasound simulations. Because the devices have their limitations and benefits regarding realism, costs, and access, we compared three technical setups: desktop with mouse interaction, desktop with a haptic device, and virtual reality with a haptic device. In a user study with 19 participants we investigated the usability and assessed qualitative user feedback in a semi-structured interview. Significant differences regarding usability and training time were found between both versions with the haptic device and the desktop with mouse version. Based on the feedback, using a haptic device seems to be more relevant than the two output devices in this training case.

Index Terms: Virtual Reality, Medical Training, Ultrasound, Haptics, Input Device.

1 INTRODUCTION

Ultrasound (US) is a real-time imaging technique that is used, for example, for abdominal examination, needle-based interventions, or during surgeries in various medical fields. The main mental challenge during this procedure is to build a mental model of the anatomy [8, 20]. Based on the US image and the transducer's position on the anatomy, the physician has to create a mental representation of the interior structures to understand them and to orient within the anatomy. The required visuospatial skill can only be

trained hands on. In case of intraoperative US (IOUS), hands-on training during surgery is difficult. But also for other applications of US, there might be ethical concerns regarding training on the patient. Because of this, virtual simulations were developed to train the required mental skills in a safe environment.

Existing US simulations (see Tab. 1) differ with respect to their output and input device. The input devices range from mouse [18] over VR controllers [4, 11, 25] to haptic devices [1, 2, 3, 7, 12, 15, 19, 21, 26]. Only the haptic device provides haptic feedback when colliding with an object and enables the user to physically sense resistance. In this case the vibration feature of VR controllers is excluded from being classified as true haptic feedback as it does not provide resistance. Beyond their feedback capabilities, these devices also differ in terms of their range of motion. A traditional computer mouse can only be moved on a planar surface, simulating two movement directions. In contrast, both the VR controller and the haptic device operate mid-air, allowing complex three-dimensional manipulation in space. The output devices range from normal monoscopic displays over stereoscopic displays to immersive virtual reality (VR). From these three devices, monoscopic displays are the only ones that cannot provide the impression of three-dimensional depth. Stereoscopic displays create a more convincing illusion of depth, whereas an immersive VR system can mirror the real arrangement of objects that surround the user. For instance, in the context of an US simulation, an immersive VR setup can accurately place key elements like the virtual patient and the ultrasound monitor. According to an observation of a real surgical procedure, the virtual patient is positioned next to the user, while the US monitor is placed higher and on the opposite side of the patient. Of course there are slightly varying setups. Nevertheless, this realistic configuration demands the same hand-eye coordination that is required in real-life medical procedures.

Accordingly, the most realistic combination is the haptic device with VR. However, desktop and mouse may facilitate broader user access and may enable the target group to use it from various places

*e-mail: mareen@isg.cs.uni-magdeburg.de

†e-mail: hansen@isg.cs.uni-magdeburg.de

Table 1: US Simulations and their input and output devices.

	Input Device			Output Device		
	Mouse	VR Controller	Haptic Device	Desktop	Stereoscopic Display	VR
Ni [19]			X	X		
Barnouin [2]			X	X		
Bublak [3]			X (US device)			X
Johnson [12]			X		X	
Shenoy [25]		X				X
Jacobsen [11]		X				X
Simon [26]			X			X
Chuan [7]			X			X
Law [15]			X	X		
Mayer [18]	X			X		
Byl [4]		X				X
Olgers [21]			X (printed probe)	X		
Allgaier [1]			X			X

due to their lower costs. For US training, a realistic movement of the probe can be considered as more relevant for the training effect than an immersive environment due to muscle memory. Accordingly, a possible trade-off could be a haptic device in combination with a desktop. Because of this trade-off, we compared the three different technical setups for an US simulation that are shown in Fig. 1.

2 VIRTUAL ULTRASOUND TRAINING

For the three technical setups, we used the scenario of IOUS for liver surgery. This means, US is performed on the opened patient during surgery to locate lesions such as tumors and to assess their positions in relation to the vasculature [17]. In case of liver surgery, there are immersive VR systems for planning [5, 13, 22, 23] as well as training and education [6, 9, 16, 24]. Besides anatomical education and laparoscopic training, Allgaier et al. [1] propose a VR-based US training application using a haptic input device. By presenting four scenarios, they focus on training the creation of a mental model when scanning the liver.

Our three versions use the texture-based US simulation from Allgaier et al. [1] and an adopted environment imitating an operating room from Huber et al. [10]. The virtual patient is approximated by a torso [14]. In the following, the technical setup of the versions are described in more detail.

V1: Haptic and VR In this version, the virtual US probe directly replicates the movements of the haptic device. When the probe collides with the liver, the user experiences haptic feedback. For providing haptic feedback, we utilized the *Haptics Direct for Unity V1* from *3D Systems*. A haptic material with a stiffness value of 0.073 was applied to the liver surface, a parameter that was established in collaboration with the clinical development team. Similar to a real operating room, the US monitor is positioned on the opposite side of the patient (see Fig. 2). This position is inspired by the operating room of our clinical partners, however, the exact position can vary in other operating rooms. Independent of the exact position, the user has to alternately shift their focus between the patient and the monitor. This challenge regarding hand-eye-coordination replicates the real scenario.

As shown in Tab. 1, there are also simulations that use VR controllers. However, there are two limitations of VR controllers in this scenario: Firstly, it is essential that the input device is moved on a support surface, either simulated by a haptic device or a physical surface, in order to provide the user with feedback on their positioning on the liver surface. During the development, liver surgeons also emphasized that a supporting surface is necessary to properly navigate a 3D input device on the liver surface. Using a VR controller would also impact the implementation of the virtual probe: When moving the controller in a way that would penetrate the liver (because there is no physical resistance), should the virtual probe collide with the virtual liver or should it also penetrate the liver? Both of these scenarios lack realism and could potentially impair the user experience. Secondly, the controller's grip differs significantly from that of an IOUS probe, which may result in the development of incorrect muscle memory after an extended training period. The mouse's grip is also bit different, however, still closer to an IOUS probe than the power grip of VR controllers.

V2: Haptic and desktop The interaction with the haptic device is the same as in V1. Because the user can freely move within the VR environment, we divided the desktop application into an exploration mode and US mode. At the beginning, the user can freely move through the virtual operating room for exploration by using the keyboard. Before starting scanning the liver, the user's position and view are fixated to properly see the liver and to move the US probe. Another difference between VR and desktop is the spatial arrangement of the patient and US monitor. In the VR environment, the user experiences a realistic setup, which requires to physically look up from the patient to the monitor. In contrast, the desktop versions show the patient and the US monitor onto a single display (see Fig. 2). It would be possible to use two displays, one showing the probe and patient and the other one displaying the US images. However, we decided to avoid this more complex setup. In contrast to a VR environment, the input device representing the US probe is not at the position of the virtual probe shown on the display. Consequently, with an additional screen, there would be two displays and the real device where the user might look at.

V3: Mouse and desktop In contrast to the haptic device, the mouse has a significant restriction of movement, making it challenging to implement similar control mechanics for the virtual probe. To address this limitation, the virtual probe is automatically oriented orthogonal to the liver surface, simplifying the interaction. This automatic alignment ensures that the probe maintains a consistent and stable orientation relative to the liver, facilitating smoother navigation and reducing the complexity of manual adjustments. To provide a rotation of the probe around its up-axis, we employ the mouse wheel. However, due to the fixation of the probe to the liver surface, tilting the probe in other directions is not possible. Although this limitation differs from the behavior of a real US probe, it simplifies the interaction for users of the mouse. Otherwise, for example, additional keyboard buttons would be necessary which is why we decided to keep the cognitive burden low by excluding these rotations.

To evaluate and compare the three versions of the system, we designed a task that demands the user to develop a good mental model and a comprehensive understanding of the internal anatomical structures of the liver. In this task, the user has to identify the number of tumors present within the liver model. This task was chosen not only for its necessity of spatial reasoning and probe manipulation skills, but also because it is clinically relevant. As mentioned before, US is routinely used during liver surgeries to detect and accurately locate all tumors, making this task a simplified version of real-world medical scenarios. To have an appropriate number of tasks, we created a pool of ten livers. We used a segmented patient liver from our medical partners and removed

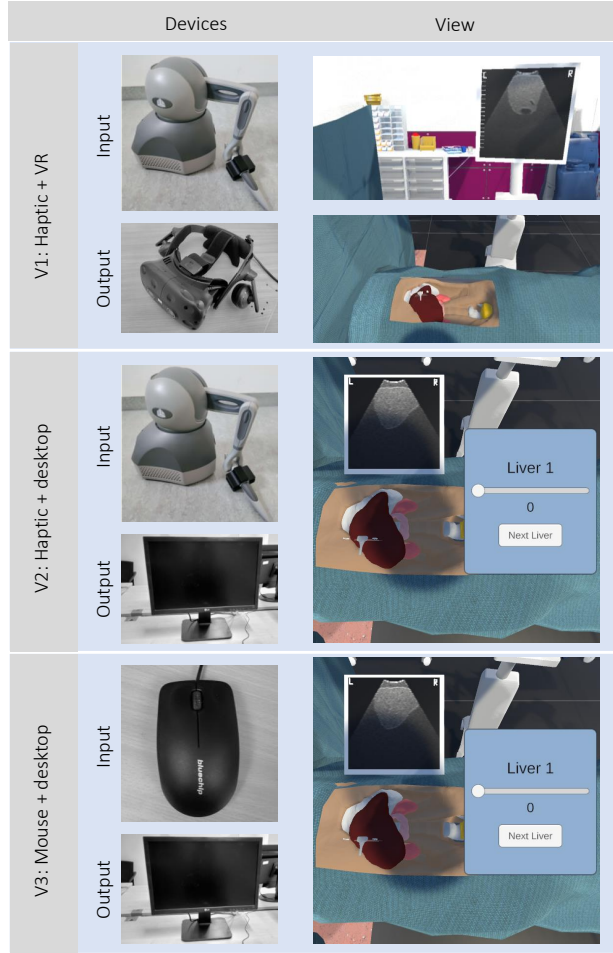


Figure 2: The three versions with their devices and the user's view.

the original tumor and metastases. Afterwards we approximated tumors by adding spheres. Thus, each liver differs solely in the number and spatial arrangement of tumors, while other anatomical features remain consistent. The number of tumors in these models ranges from zero to four.

3 USER STUDY

3.1 Apparatus

For the version using VR, the *HTC Vive Pro Eye* (HTC Corporation, Taiwan) was used. All versions using a haptic device employ the *Geomagic Touch* (3D Systems, United States). For the other versions, a standard desktop and mouse setup was used.

In this initial within-subject design study, we decided to evaluate the three versions with non-medical participants. This decision is based on the limited availability of medical experts for the study. By involving non-medical participants, we aimed to gather preliminary insights into the usability, functionality, and effectiveness of the systems without being constrained by participant availability. Although non-medical participants do not possess the same level of domain-specific knowledge, they can identify usability issues and give feedback on how intuitive the systems are for novice users. A total of 19 participants were included in the study, and their demographic characteristics are presented in Tab. 2.

Table 2: Characteristics of participants ($n = 19$).

Characteristics	Value	Mean
Age [years, mean (range)]	[22-29]	$\bar{x} = 26$
20-25	4	(21%)
26-30	15	(79%)
Gender		
Male	11	(58%)
Female	8	(42%)
Experience 3D input devices		
None	6	(32%)
Medium	13	(68%)
Much	0	(0%)
Experience with VR		
None	3	(16%)
Medium	16	(84%)
Much	0	(0%)

3.2 Study Procedure

The overall procedure is shown in Fig. 3. Starting with an introduction, the participants were informed about the procedure and interactions. Additionally, the introduction included a demographic questionnaire. In the main part, the participants had to repeat the previously described task with all three versions. At the beginning of each version, they had unlimited time to train scanning a training liver with the current version. After each version, the system usability scale (SUS) and NASA-TLX for load must be completed. The objective of using these two questionnaires was to obtain a general impression of the versions regarding their usability and suitability for performing US. During the tasks, errors and training time as well as time required for the tasks were recorded. After all three versions were completed, a semi-structured interview was conducted, including the following questions:

- Would you be motivated to continue with one or more of the versions or try out other types of tasks?
- Did something stand out in one of the different systems that you liked most or least?
- With which of the three versions were you able to examine the livers most effectively and which was the least effective?
- Which application do you find the most intuitive or the quickest to learn? And which one takes the most time to learn?
- In your opinion, what were the biggest challenges of the tasks? Did you notice differences between the versions?
- Which input and output devices did you find better or more suitable?
- Do you have any suggestions for improving one or more versions, in particular with regard to technical implementation?

3.3 Results and Discussion

Using one-way ANOVA, we found statistically significant differences regarding the training time and SUS (see Tab. 3). The post-hoc analysis revealed that V1 and V2 had a significantly longer training time than V3. As the NASA TLX scores show no significant differences (V1: 21, $\sigma = 10$; V2: 23, $\sigma = 15$; V3: 26, $\sigma = 13$), the difference might be explained due to a higher familiarity with mouse interactions and not due to difficulty in handling. This suggestion is also supported by the observation that despite of

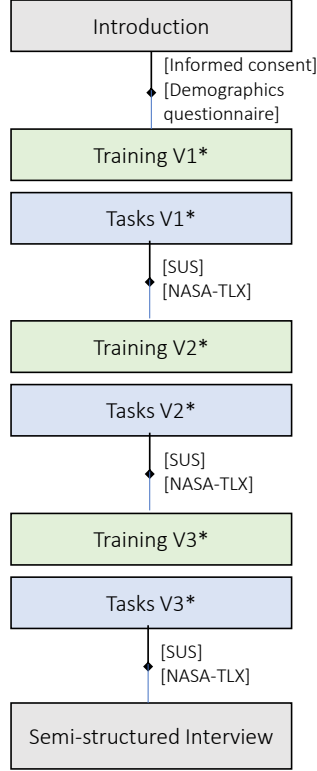


Figure 3: Procedure of the user study. * indicated randomized orders of the versions.

Table 3: Results of the one-way ANOVA. * indicates significant differences.

Variable	F	p	η^2
Training time [s]	10.71	0.0001*	0.284
Task time [s]	1.63	0.2058	0.057
Success rate [%]	1.85	0.1675	0.064
SUS	5.92	0.0048*	0.180
NASA TLX	0.54	0.5884	0.019

the shorter training time, the time for the task is not significantly different. Regarding the SUS, V1 and V2 also performed significantly better than V3. The descriptive results are summarized in Tab. 4. Looking at the single items of the NASA TLX, it can be seen that V1 has a higher mental and physical load, however, it achieves the best result regarding performance, effort and frustration followed by V2.

The qualitative feedback of the participants was collected and categorized. Regarding functionality, the participants stated that the shape of the liver can be better recognized in VR, whereas the desktop application has the benefit that everything is visible on one screen. Comparing the two input devices, they recognized that the mouse provides less input and control possibilities and the haptic feedback of the haptic device supports the orientation on the liver. However, some mentioned that the 3D haptic device is only more effective than the mouse because of the haptic feedback and not because of the freedom of movement.

Statements concerning usability comprise that VR is more fun but it is also perceived as more burdensome. Some participants stated looking up and down between the patient and monitor is exhausting. Because this reflects the real situation, medical experts

Table 4: Quantitative results of the user study. * indicates significant differences.

Variable	V1	V2	V3
Training time [s]*	01:09±00:20	00:59±00:22	00:41±00:12
Task time [s]	04:18±01:19	05:01±01:36	04:59±01:12
Success rate [%]	83±19	79±26	69±21
SUS*	79±17	81±15	62±22
NASA TLX	21±10	23±15	26±13

might assess this differently. Regarding the input devices, the participants have the impression that the haptic device is better for learning and less exhausting comparing to mouse interaction.

With respect to intuitiveness, participants had different opinions. Ten participants mentioned that mouse interactions are familiar and easy to learn. However, four participants mentioned that the haptic device is the most intuitive one. Eight participants differentiated by stating that the haptic control is more difficult to learn but after a short time it is superior to mouse interaction. These statements are in accordance with the significantly shorter training time of the mouse device and the similar task time.

In case of realism, the participants stated that the VR simulation and the haptic device feel more realistic, whereas the desktop application feels more like a game.

In conclusion, the results show that the haptic device requires more time to familiarize with, but this has no negative effect on performance or task duration. Furthermore, the versions with haptic device reached a higher SUS score. Because there are only significant differences between the versions with haptic device and the version with mouse interaction and based on the qualitative feedback, the input device seems to be more important than the output device.

After gaining first insights with the presented study, it is necessary to conduct a follow-up study involving medical experts. Medical experts as participants can be divided into two groups. Firstly, surgeons who have experience with IOUS should qualitatively assess the three versions, for example, regarding the overall realism, monitor and patient positioning, and the resulting hand-eye-coordination. Secondly, a study with novice liver surgeons with little or no practical experience with IOUS, representing the target group, should be conducted to investigate the learning effect using the three versions. Even if one version has a lower usability or is less realistic, it could lead to a similar training effect and might benefit from, for example, accessibility. Thereby, it is important to assess the learning transfer, which means whether the learned skills can be transferred to the operating room, and not the learning effect within the training environment. This study can differentiate between several learning objectives such as mental skills or probe manipulation.

REFERENCES

- [1] M. Allgaier, F. Huettl, L. I. Hanke, T. Huber, B. Preim, S. Saalfeld, and C. Hansen. LiVRSono-virtual reality training with haptics for intraoperative ultrasound. In *Proc. ISMAR*, pp. 980–989. IEEE Computer Society, Los Alamitos, 2023. doi: 10.1109/ISMAR59233.2023.00114 1, 2
- [2] C. Barnouin, F. Zara, and F. Jaillet. A real-time ultrasound rendering with model-based tissue deformation for needle insertion. In *Proc. GRAPP*, pp. 235–246. SciTePress, Setúbal, 2020. doi: 10.5220/0008947302350246 1, 2
- [3] T. Bublak, M. Bofferding, C. Olson, H. Jonas, J. Rademeier, and C. Hansen. A virtual environment for emergency ultrasound training during cardiopulmonary resuscitation. In *Proc. MuC*, pp. 600–602. Association for Computing Machinery, New York, NY, USA, 2022. doi: 10.1145/3543758.3547516 1, 2

- [4] B. Byl, M. Sünksen, and M. Teistler. A serious virtual reality game to train spatial cognition for medical ultrasound imaging. In *Proc. SeGAH*, pp. 1–4, 2018. doi: 10.1109/SeGAH.2018.8401365 1, 2
- [5] V. Chheang, R. Bruggemann, B. Preim, and C. Hansen. Virtual resection planning using bezier surface interactions in collaborative vr environments. In *Proc. VRW*, pp. 166–169. IEEE Computer Society, Los Alamitos, 2023. doi: 10.1109/VRW58643.2023.00041 2
- [6] V. Chheang, P. Saalfeld, T. Huber, F. Huettl, W. Kneist, B. Preim, and C. Hansen. Collaborative virtual reality for laparoscopic liver surgery training. In *Proc. AIVR*, pp. 1–17. IEEE Computer Society, Los Alamitos, 2019. doi: 10.1109/AIVR46125.2019.00011 2
- [7] A. Chuan, J. Qian, A. Bogdanovych, A. Kumar, M. McKendrick, and G. McLeod. Design and validation of a virtual reality trainer for ultrasound-guided regional anaesthesia. *Anaesth*, 78(6):739–746, 2023. doi: 10.1111/anae.16015 1, 2
- [8] E. J. Hagopian. Liver ultrasound: A key procedure in the surgeon’s toolbox. *J Surg Oncol*, 122(1):61–69, 2020. doi: 10.1002/jso.25908 1
- [9] T. Huber, M. Paschold, C. Hansen, T. Wunderling, H. Lang, and W. Kneist. New dimensions in surgical training: immersive virtual reality laparoscopic simulation exhilarates surgical staff. *Surg Endosc*, 31:4472–4477, 2017. doi: 10.1007/s00464-017-5500-6 2
- [10] T. Huber, T. Wunderling, M. Paschold, H. Lang, W. Kneist, and C. Hansen. Highly immersive virtual reality laparoscopy simulation: Development and future aspects. *Int J Comput Assist Radiol Surg*, 13:281–290, 11 2017. doi: 10.1007/s11548-017-1686-2 2
- [11] N. Jacobsen, J. D. Larsen, C. Falster, C. P. Nolsøe, L. Konge, O. Graumann, and C. B. Laursen. Using immersive virtual reality simulation to ensure competence in contrast-enhanced ultrasound. *Ultrasound Med Biol*, 48(5):912–923, 2022. doi: 10.1016/j.ultrasmedbio.2022.01.015 1, 2
- [12] S. Johnson, C. Hunt, H. Woolnough, M. Crawshaw, C. Kilkenny, D. Gould, A. Sinha, A. England, and P.-F. Villard. Virtual reality, ultrasound-guided liver biopsy simulator: Development and performance discrimination. *Br J Radiol*, 85, 02 2011. doi: 10.1259/bjr/47436030 1, 2
- [13] H. G. Kennigott, M. Pfeiffer, A. A. Preukschas, L. Bettscheider, P. A. Wise, M. Wagner, S. Speidel, M. Huber, F. Nickel, A. Mehrabi, and B. P. Müller-Stich. Imhotep: cross-professional evaluation of a three-dimensional virtual reality system for interactive surgical operation planning, tumor board discussion and immersive training for complex liver surgery in a head-mounted display. *Surg Endosc*, 36(1):126–134, 2022. doi: 10.1007/s00464-020-08246-4 2
- [14] H. G. Kennigott, J. J. Wünscher, M. Wagner, A. Preukschas, A. L. Wekerle, P. Neher, S. Suwelack, S. Speidel, F. Nickel, D. Oladokun, L. Maier-Hein, R. Dillmann, H. P. Meinzer, and B. P. Müller-Stich. Openhelp (heidelberg laparoscopy phantom): development of an open-source surgical evaluation and training tool. *Surg Endosc*, 29(11):3338–3347, 2015. doi: 10.1007/s00464-015-4094-0 2
- [15] Y. C. Law, T. Knott, S. Pick, B. Weyers, and T. W. Kuhlen. Simulation-based Ultrasound Training Supported by Annotations, Haptics and Linked Multimodal Views. In *Proc. VCBM*. The Eurographics Association, Eindhoven, 2015. doi: 10.2312/vcbm.20151220 1, 2
- [16] A. K. Lefor, S. A. Heredia Pérez, A. Shimizu, H. C. Lin, J. Witowski, and M. Mitsuishi. Development and validation of a virtual reality simulator for robot-assisted minimally invasive liver surgery training. *J Clin Med*, 11:4145, 2022. doi: 10.3390/jcm11144145 2
- [17] M. G. Lubner, L. Mankowski Gettle, D. H. Kim, T. J. Ziemlewicz, N. Dahiya, and P. Pickhardt. Diagnostic and procedural intraoperative ultrasound: technique, tips and tricks for optimizing results. *J Radiol*, 94(1121):20201406, 2021. doi: 10.1259/bjr.20201406 2
- [18] L. Mayer, M. Sünksen, S. Reinhold, S. Bertel, and M. Teistler. Training visuospatial skills for medical ultrasound imaging with a desktop-based learning game. In *Proc. SeGAH*, pp. 1–6, 2021. doi: 10.1109/SeGAH52098.2021.9551914 1, 2
- [19] D. Ni, W. Y. Chan, J. Qin, Y.-P. Chui, I. Qu, S. S. M. Ho, and P.-A. Heng. A virtual reality simulator for ultrasound-guided biopsy training. *IEEE Comput Graph Appl*, 31(2):36–48, 2011. doi: 10.1109/MCG.2009.151 1, 2
- [20] D. Nicholls, L. Sweet, and J. Hyett. Psychomotor skills in medical ultrasound imaging. *J Med Ultrasound*, 33(8):1349–1352, 2014. doi: 10.7863/ultra.33.8.1349 1
- [21] T. Olgers, J. Os, H. Bouma, and J. Maaten. The validation of a serious game for teaching ultrasound skills. *Ultrasound J*, 14(29), 2022. doi: 10.1186/s13089-022-00280-8 1, 2
- [22] A. V. Reinschluessel, T. Muender, D. Salzmann, T. Döring, R. Malaka, and D. Weyhe. Virtual reality for surgical planning - evaluation based on two liver tumor resections. *Front Surg*, 9:821060, 2022. doi: 10.3389/fsurg.2022.821060 2
- [23] B. Reitingner, A. Bornik, R. Beichel, and D. Schmalstieg. Liver surgery planning using virtual reality. *IEEE Comput Graph Appl*, 26:36–47, 2006. doi: 10.1109/MCG.2006.131 2
- [24] D. Schott, P. Saalfeld, G. Schmidt, F. Joeres, C. Boedecker, F. Huettl, H. Lang, T. Huber, B. Preim, and C. Hansen. A vr/ar environment for multi-user liver anatomy education. In *Proc. VR*, pp. 296–305. IEEE Computer Society, Los Alamitos, 2021. doi: 10.1109/VR50410.2021.00052 2
- [25] V. Shenoy, S. I. Khan, E. Lee, and O. Aalami. Vascular1: Development and evaluation of a virtual reality ultrasound guided vascular access training module. In *Proc. BIBM*, pp. 1519–1523. IEEE Computer Society, Los Alamitos, 2021. doi: 10.1109/BIBM52615.2021.9669886 1, 2
- [26] C. Simon, L. Herfort, and A. Chellali. Design and evaluation of an immersive ultrasound-guided locoregional anesthesia simulator. In *Proc. VRW*, pp. 445–449. IEEE Computer Society, Los Alamitos, 2022. doi: 10.1109/VRW55335.2022.00098 1, 2